

The Evolution of Cooperation

Term Project

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Abstract

This report studies how cooperation emerges in a society of selfish actors. Cooperation is modeled as interactions between players of the Prisoner's Dilemma. Players are modeled with various reactive strategies using Asymmetric Cell-DEVS and implemented Cadmium. Two experiments were run, both showing the resilience of the Tit for Tat (TFT) strategy in asymmetric cell spaces.

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1. Introduction

We don't know whether human nature is selfish or generous, egoist or altruistic. We don't know because we can't observe our pre-social nature; we can't strip ourselves of society, rules, conventions, and religions. Taking the pessimistic assumption that we are all inherently selfish, how then do we account for behaviour such as generosity and forgiveness? How does cooperation emerge in a society of egoists?

We can model cooperation between two selfish agents using the Prisoner's Dilemma, a widely studied model in game theory. In the classic version of this model, it has been shown that for a single game the best move for either agent is to not cooperate, i.e., to be selfish. However, for consecutive games between players, we observe the best strategy is to cooperate with one another some of time. We can use this model to study how cooperation emerges in a society of selfish actors.

In Axelrod's Tournament [1], contestants were asked to submit computer algorithms to compete against one another in consecutive games of the Prisoner's Dilemma. The best strategy, the one that accumulated the most points, was shown to be Tit for Tat (TFT), where a player would cooperate if their opponent cooperated on the previous turn or they would not cooperate if their opponent did not cooperate on the previous turn. Patrick Grim [2] extended Axelrod's tournament to a spatialized format, modeling players as cellular automata in a 2D matrix. The tournament was initialized with players with different strategies distributed randomly throughout the cell-space. After each round, players would adopt the best performing strategy of their neighbours and play again. TFT, again, proved victorious.

But what happens if we change the cell space? Our universe and interactions are not grid-like, they are not symmetrical. That is, for any given day, I don't interact with the same number of people as my neighbour, nor would someone living in a rural community interact with the same number of people as someone living in a city. The questions we want to answer then are:

- How do an asymmetrical cell spaces affect the results of Axelrod's Tournament?
- Can inferior strategies in highly connected players dominate the better strategies of more isolated players?

In this report we model and simulate Axelrod's Tournament in an asymmetric cell space and study how cooperation emerges amongst selfish actors. I implement the spatialized Prisoner's Dilemma model with the reactive strategies studied in [2] using Asymmetric Cell-DEVS. I will validate the model by reconstructing the symmetrical cell space studied in [2] and confirming that TFT is the dominant strategy. I will then modify the cell space to be asymmetric and run experiments to see what conditions result in different strategies emerging dominant.

Section 2 describes background information. Section 3 defines the conceptual and Asymmetric Cell-DEVS model. Section 4 presents and discusses the simulation results. Lastly, Section 5 concludes the report.

2. Background

2.1. The Prisoner's Dilemma

The Prisoner's Dilemma is a widely studied model in game theory. When implemented as cellular automata, the model is referred to as the Spatialized Prisoner's Dilemma and can be used to study how cooperation emerges in a society of selfish actors [2].

The Prisoner's Dilemma describes two prisoners each of whom can "cooperate" for mutual benefit or "defect" for personal gain. The classic version of the game describes two prisoners detained for a crime they jointly committed and put in separate rooms where they can't communicate with each other. The prosecutor only has enough evidence to send them each to prison for two years. So, the prosecutor offers each prisoner a deal. If a prisoner defects (i.e., snitches on their partner), then they walk free and the other goes to prison for five years. However, if they both defect, then they both go to prison for four years. Table 1 represents the prisoner's payoffs (i.e., the number of years they are free) as a matrix. This matrix is commonly parameterized to be the reward (R), sucker's (S), temptation (T), and punishment (P) payoffs as shown in Table 2.

What should each prisoner do? Ideally, both sides should cooperate. But in this version, if Player A wants to cooperate, Player B has a better payoff if they choose to defect. And if Player A wants to defect, Player B still has a better payoff if they choose to defect. Thus, Player B should always defect. This argument is symmetrical for Player A. Here we see that it is better for both sides to be selfish, i.e., to defect. We refer to prisoners as players from here on.

Table 1: Classic payoff matrix.

		Player A	
		Cooperate	Defect
Player B	Cooperate	3, 3	0, 5
	Defect	5, 0	1, 1

Table 2: Parameterized payoff matrix.

		Player A	
		Cooperate	Defect
Player B	Cooperate	R, R	S, T
	Defect	T, S	P, P

But what happens when we play consecutive games of the Prisoner's Dilemma? Both sides could continually defect, but to maximize each player's payoff they should cooperate with each other some of the time. This is an example of cooperation emerging in a society of selfish actors.

In consecutive games, players can now have a variety of *reactive strategies* where their next move depends on their opponent's previous move. As shown in [2], if we encode behaviour as integers, where defection is the integer zero and cooperation is the integer one, we can exhaustively list all the strategies. Table 3 describes these strategies, where:

- i represents the player's initial move
- c represents how the player reacts to an opponent cooperating
- d represents how the player reacts to an opponent defecting

Table 3: List of reactive strategies.

i	c	d	<i>reactive strategy</i>
0	0	0	Defector
0	0	1	Suspicious doormat
0	1	0	Suspicious Tit for Tat
0	1	1	Suspicious cooperator
1	0	0	Deceptive defector
1	0	1	Gullible doormat
1	1	0	Tit for Tat (TFT)
1	1	1	Cooperator

2.2. Asymmetric Cell-DEVS

Cell-DEVS combines the Discrete Event System Specification (DEVS) formalism and cellular automata (CA) formalism. Asymmetric Cell-DEVS is a generalization of Cell-DEVS to complex topologies [3]. In contrast to traditional cellular models, asymmetric cell models allow neighbourhood sizes and types to be different for every cell. Asymmetric Cell-DEVS models are defined as:

$$CD = \langle X, Y, C, EIC, IC, EOC \rangle$$

where:

- X is the input set.
- Y is the output set.
- C is the subcomponent set. It defines all the cells in the model.
- EIC is the external input coupling set. It maps X to inputs of cells.
- IC is the internal coupling set. It defines how the cells are connected.
- EOC is the external output coupling set. It maps outputs of cells to Y .

For each cell $C_i \in C$, the cell model is defined as:

$$C_i = \langle X_i^N, X_i^E, Y_i, S_i, N_i, \tau_i, d_i, delay_i \rangle$$

where:

- X_i^N is the neighbourhood input set. It defines the possible inputs from neighbouring cells.
- X_i^E is the external input set. It defines the inputs a cell can receive that do not originate from a neighbouring cell.
- Y_i is the cell output set.
- S_i is the cell state set.
- N_i is the neighbourhood set. It lists all the neighbouring cells.
- $\tau_i: S_i \times N_i \times X_i \rightarrow S_i$ is the local computation function of the cell. It computes the new state of cell C_i from its previous state, the neighbourhood set, and inputs.
- $d_i: S_i \rightarrow \mathbb{R}_{\geq 0}$ is the cell delay function. It computes how long the cell should wait after a state change before sharing with other cells that it has changed.
- $delay_i$ is the cell delay type function.

3. Models

3.1. Conceptual model

Each cell is a player in the Prisoner's Dilemma. Each player has a reactive strategy from Table 3. Players are arranged in an asymmetric cell space and play rounds of the Prisoner's Dilemma with each of their neighbours. Players accumulate payoff as defined in Table 1 and track which of their neighbours has the highest payoff.

3.2. Formal specification

Let *Strategies* be the set of all reactive strategies defined in Table 3. Let variables R, S, T, P respectively be the reward, sucker's, temptation, and punishment payoffs from the parameterized payoff matrix in Table 2.

Table 4 describes the cell model for $C_i \in C$. For clarity, the definition focuses on the cell's state variables (S_i) and the local computing function (τ_i), which involves playing a round of the Prisoner's Dilemma with each of the cell's neighbours. The cell's state variables are:

- *moves* is a map of cell IDs to moves. A move is how the player will act in the next round and is *True* when the player cooperates and *False* when the player defects.
- *total_payoff* is the sum of all the payoffs the player received throughout all the rounds of a tournament.
- *strategy* is the strategy this cell is following.

Table 4: Cell definition for a player.

$S =$	$\{ \text{moves}: \mathbb{Z}^+ \rightarrow \text{Bool},$ $\text{totalPayoff} \in \mathbb{Z}^+,$ $\text{strategy} \in \text{Strategies} \}$
$\tau =$	<pre> for n in N: // Get previous moves from each player a ← self.moves.get(n.id); b ← n.moves.get(self.id); // Play the Prisoner's Dilemma else if a = True and b = True: totalPayoff ← totalPayoff + R; else if a = True and b = False: totalPayoff ← totalPayoff + S; else if a = False and b = True: totalPayoff ← totalPayoff + T; else: totalPayoff ← totalPayoff + P; // Decide next move based on opponent's move m ← strategize(b, strategy); self.moves.update(n.id, m); </pre>

3.3. Model verification

The Asymmetric Cell-DEVS model in

Table 4 was implemented in Cadmium. The test suite was designed to minimize our test cases while ensuring we test each strategy at least once. We need to test more than one round to verify that both the initial move and how the player reacts are correct, so each test consists of 200 rounds as per Axelrod's Tournament [1]. As calculated in [2], Table 5 shows the total payoff for the reactive strategies in competition with each other for 200 rounds. Table 6 shows the chosen test suite with the strategy pairings and their expected results. Figure 1, Figure 2, Figure 3, and Figure 4 show the results of the tests.

Table 5: Matrix of total payoffs for the reactive strategies in competition with each other for 200 rounds.

	$\langle 0,0,0 \rangle$	$\langle 0,0,1 \rangle$	$\langle 0,1,0 \rangle$	$\langle 0,1,1 \rangle$	$\langle 1,0,0 \rangle$	$\langle 1,0,1 \rangle$	$\langle 1,1,0 \rangle$	$\langle 1,1,1 \rangle$
$\langle 0,0,0 \rangle$	200	996	200	996	204	1000	204	1000
$\langle 0,0,1 \rangle$	1	400	450	994	6	1000	450	1000
$\langle 0,1,0 \rangle$	200	450	200	600	203	450	500	602
$\langle 0,1,1 \rangle$	1	4	595	598	5	8	599	602
$\langle 1,0,0 \rangle$	199	991	203	995	202	994	206	998
$\langle 1,0,1 \rangle$	0	0	450	993	4	400	450	998
$\langle 1,1,0 \rangle$	199	450	500	599	201	450	600	600
$\langle 1,1,1 \rangle$	0	0	597	597	3	3	600	600

Table 6: Test suite.

Test case	Strategy encoding		Expected payoff	
	s_1	s_2	p_1	p_2
Defector vs. suspicious doormat	$\langle 0,0,0 \rangle$	$\langle 0,0,1 \rangle$	996	1
Suspicious TFT vs. suspicious cooperator	$\langle 0,1,0 \rangle$	$\langle 0,1,1 \rangle$	600	595
Deceptive defector vs. gullible doormat	$\langle 1,0,0 \rangle$	$\langle 1,0,1 \rangle$	994	4
TFT vs. cooperator	$\langle 1,1,0 \rangle$	$\langle 1,1,1 \rangle$	600	600

```

Test case 1: Defector vs suspicious doormat
Simulation logs: out/test_defector_vs_susp_doormat_log.csv
Expected result:
Defector: <0,0,996>
Suspicious doormat: <1,0,1>
Actual result:
199;1;defector;;<0,0,996>
199;2;susp_doormat;;<1,0,1>

```

Figure 1: Test defector vs. suspicious doormat.

```

Test case 2: Suspicious TFT vs suspicious cooperator
Simulation logs: out/test_susp_tft_vs_susp_cooperator_log.csv
Expected result:
Suspicious TFT: <2,2,600>
Suspicious cooperator: <3,2,595>
Actual result:
199;1;susp_cooperator;;<3,2,595>
199;2;susp_tft;;<2,2,600>

```

Figure 2: Test suspicious TFT vs. suspicious cooperator.

```

Test case 3: Deceptive defector vs. gullible doormat
Simulation logs: out/test_deceptive_defector_vs_gullible_doormat_log.csv
Expected result:
Deceptive defector: <4,4,994>
Gullible doormat: <5,4,4>
Actual result:
199;1;deceptive_defector;;<4,4,994>
199;2;gullible_doormat;;<5,4,4>

```

Figure 3: Test deceptive defector vs. gullible doormat.

```

Test case 4: TFT vs. cooperator
Simulation logs: out/test_tft_vs_cooperator_log.csv
Expected result:
TFT: <6,6,600>
Cooperator: <7,7,600>
Actual result:
199;1;cooperator;;<7,7,600>
199;2;tft;;<6,6,600>

```

Figure 4: Test TFT vs. cooperator.

3.4. Model validation

The model was validated by reconstructing an experiment performed by Grim [2] where the evolution of 64 x 64 toroidal array with a random configuration of reactive strategies is studied. Each cell plays 200 rounds with their neighbours and then adopts the strategy of the highest performing neighbour. After successive tournaments, the Defector strategy seems like it is going to win as the dominant strategy, however, the TFT strategy makes a comeback and proves dominant. The neighbourhood configuration was not specified in the experiment; a Von Neumann neighbourhood is assumed.

Figure 5 shows the validation results and Table 7 shows the color codes for the strategies, the Wong color palette was used to optimize for color-blind individuals [4]. After 12 simulation runs, TFT emerges as the dominant strategy.

Table 7: Strategy color codes.

Color	Strategy	Color name	RGB (0-255)
	Defector	Blue	0, 114, 178
	Suspicious doormat	Reddish purple	204, 121, 167
	Suspicious TFT	Orange	230, 159, 0
	Suspicious cooperator	Bluish green	0, 158, 115
	Deceptive defector	Sky Blue	86, 180, 233
	Gullible doormat	Black	0, 0, 0
	TFT	Yellow	240, 228, 66
	Cooperator	Vermillion	213, 94, 0
	No strategy	White	255, 255, 255

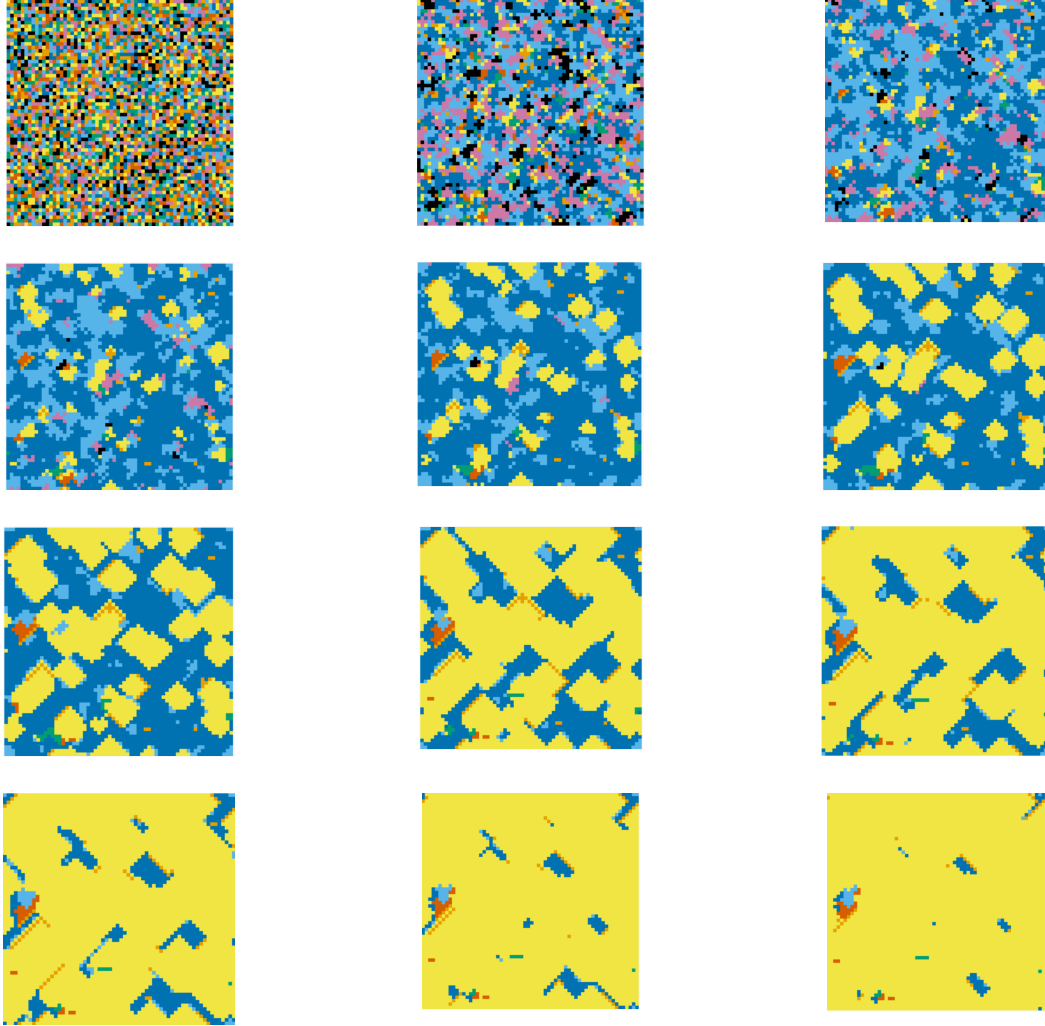


Figure 5: Validation results. 12 simulation runs (left to right, top to bottom) from random initial configuration of eight reactive strategies to a dominant TFT strategy (yellow).

4. Simulation results

The following experiments were chosen to work with the Cell-DEVS Web Viewer, which can only visualize 2D grids. Table 8 describes the experimental frame. The experiments study whether more connected players with weaker strategies can dominate less connected players with better strategies.

Table 8: Experimental frame.

Number of rounds	200, as per Axelrod's Tournament [1]
Payoff matrix	Classic (see Table 1)
Neighborhood	Von Neumann
Cell space	Toroidal 64 x 64 array
Strategies	All reactive strategies (see Table 3)

4.1. Case 1: Introduce asymmetry

Some cells were randomly populated with no strategies to create an asymmetric cell space. If a cell has no strategy, then it won't play the Prisoner's Dilemma against its neighbors, effectively reducing the size of some neighborhoods by 1 for each neighboring cell with no strategy. Strategies were randomly and evenly distributed. Figure 6 shows the simulation results. After 12 simulation runs, TFT emerges dominant.

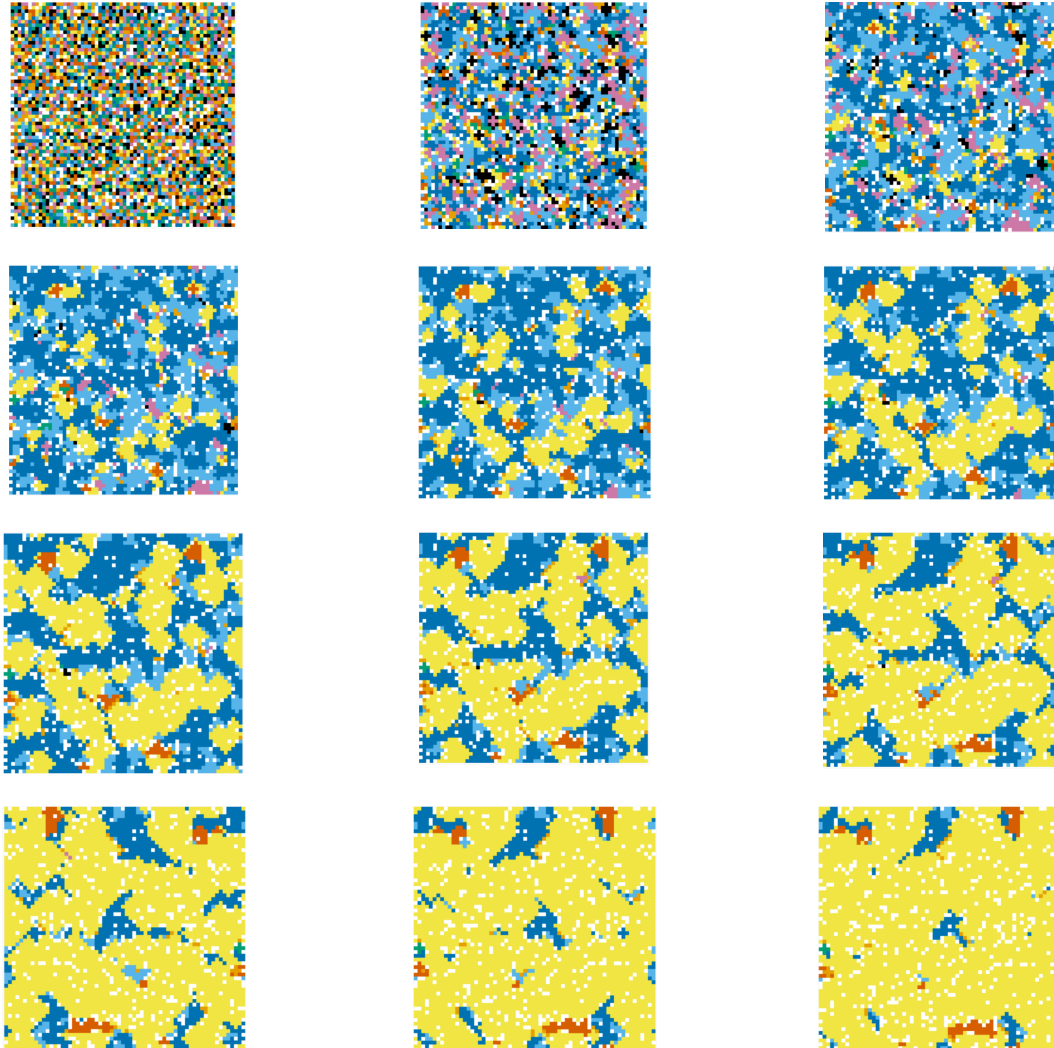


Figure 6: Case 1 simulation results. 12 simulation runs (left to right, top to bottom) from random initial configuration of eight reactive strategies. Cells with no strategies (white) are removed from neighborhoods. TFT (yellow) emerges dominant.

4.2. Case 2: Penalize TFT

For this experiment, we penalize TFT, such that if a cell is assigned the TFT strategy and its neighborhood size is greater than 4, we switch it to another random strategy. This guarantees that TFT is not the most connected player. Figure 7 shows the results of 24 simulation runs. After 24 runs, TFT appears dominant followed by Suspicious Cooperator.

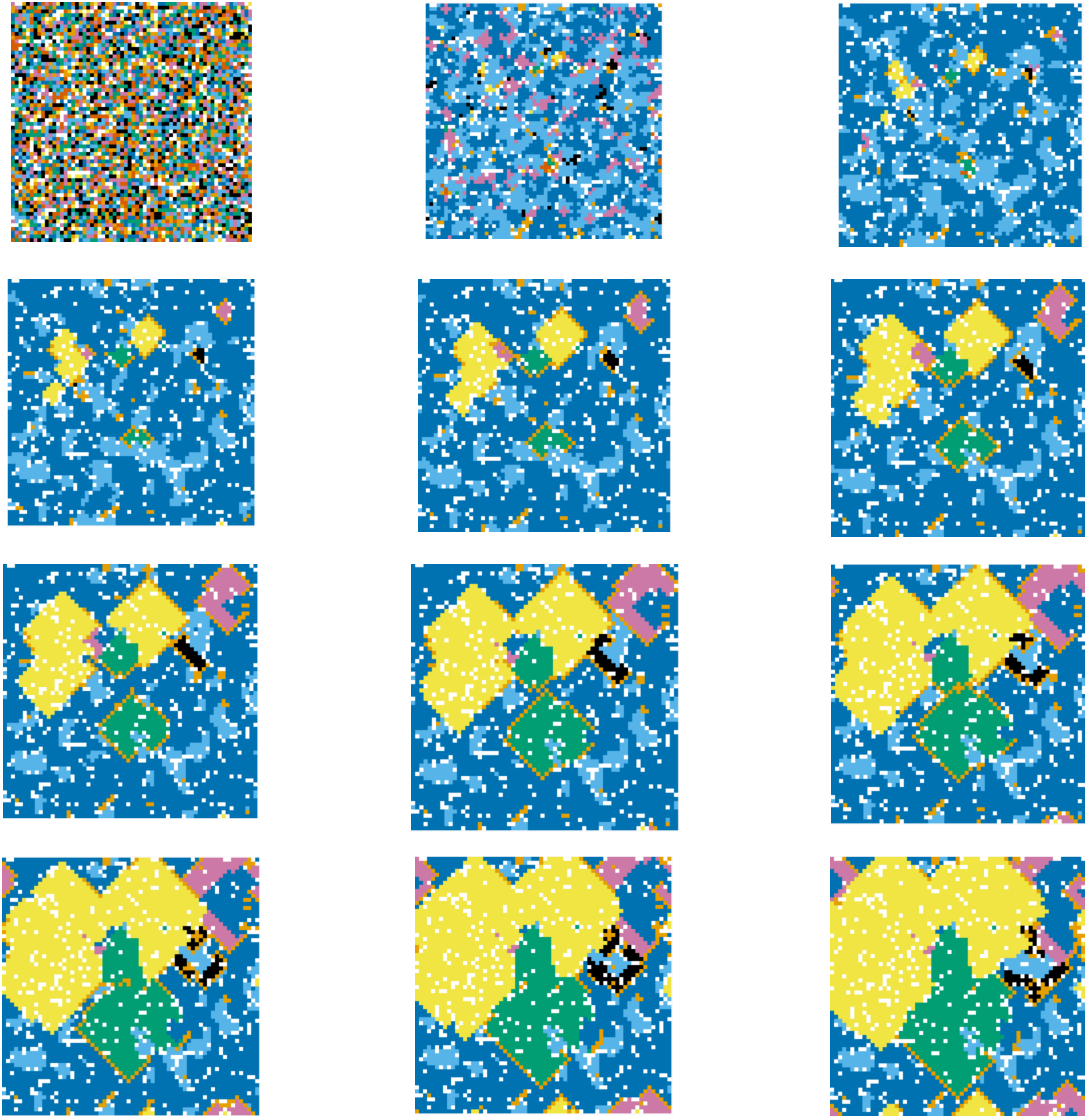


Figure 7: Case 2 simulation results. 24 simulation runs starting with a random configuration of reactive strategies with a penalty on TFT. Images show the even numbered runs (left to right, top to bottom). TFT (yellow) appears dominant followed by Suspicious Cooperator (bluish green).

4.3. Discussion

Case 1 simulation results don't differ much the validation results. However, case 2 simulation results show how resilient TFT is despite the penalty in neighborhood size and representation, since cells with a large neighborhood and a TFT strategy had their strategy replaced with another random strategy. An unexpected result is the strength of the Suspicious Cooperator (bluish green) strategy, which covers a substantial area, although never surpassing TFT. It's also interesting to note how Suspicious TFT (orange) acts as a frontline for the Suspicious Cooperator and Cooperator (vermillion) strategies in their expansion into Defector (blue) territory.

5. Conclusion

This report studied how cooperation emerges in a society of selfish actors. We did this by modeling players of the Prisoner's Dilemma as cellular automata in an asymmetric cell space using Asymmetric Cell-DEVS. Models were implemented in Cadmium, and results were visualized using the Cell-DEVS Web Viewer. Source code for the project, simulation results, and animations of select simulation runs are publicly available at [5].

Future work consists of the following. First, having a payoff matrix where its values are dependent on the neighborhood size. A constant payoff matrix is of no concern in a symmetrical cell space where every cell has the same neighborhood size, however, considerations should be made as to whether payoff should be relative to a cell's neighborhood size in an asymmetric cell space. Second, introducing stochastics and imperfect information to model forgiveness and generosity. The Asymmetric Cell-DEVS vicinity factor may be a useful way to implement this. Third, simulating asymmetric cell spaces that reflect the real-world, such as ridings in election districts or provinces within a country.

References

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